1 Randomized Least Squares

Randomized numerical linear algebra uses randomness as a computational resource for restructuring expensive linear algebra tasks into smaller or more informative surrogate problems. The area now covers low-rank approximation, matrix decompositions, regression, and randomized preconditioning, but overdetermined least-squares problems remain one of its clearest and most influential successes [1] [2] [3].

This chapter focuses on sketching-based least-squares methods, which provide the first model problem for the thesis. The starting point is the overdetermined problem

$$\min_{x\in\mathbb{R}^d} \|Ax - b\|_2,$$

where $A \in \mathbb{R}^{n\times d}$ has many more rows than columns. Interpreted geometrically, one seeks the point in the range of the linear map represented by A that is closest to b. When n is large, the dominant cost lies not in the abstract existence theory, but in the repeated dense linear algebra required to solve or approximate the system.

1.1 Sketch-and-Solve Principle

The basic randomized idea is to replace the original problem by a smaller one. Given a sketching matrix $S \in \mathbb{R}^{s\times n}$ with $s \ll n$, one solves

$$\min_{x\in\mathbb{R}^d} \|SAx - Sb\|_2.$$

If S approximately preserves the geometry of the subspace carrying the least-squares information, then the reduced problem can deliver a solution whose residual is close to optimal at a lower computational cost. In operator language, the sketch should act almost isometrically on the column space of A, or more cautiously on the augmented subspace associated with both A and b. This subspace-embedding viewpoint is one of the organizing principles of modern RandNLA [3] [2].

The importance of this formulation is that it separates two questions that are easy to conflate. The first is approximation: does the sketch preserve enough geometry for the reduced problem to represent the original one faithfully? The second is computation: can the sketch be formed and used quickly enough to justify replacing the classical solver? Early least-squares algorithms already showed that this tradeoff can be favorable in practice [4] [5].

1.2 Subspace Embeddings

The geometric engine behind sketching is the notion of an oblivious subspace embedding (OSE). The definition is simple but its consequences run deep.

Definition 1.1 (Oblivious Subspace Embedding). Let $\epsilon \in (0,1)$ and $\delta \in (0,1)$. A distribution $\mathcal{S}$ over $\mathbb{R}^{s\times n}$ is a (d,ϵ,δ)-oblivious subspace embedding if, for every fixed subspace $\mathcal{V} \subseteq \mathbb{R}^n$ of dimension at most d,

$$\Pr_{S\sim\mathcal{S}}\Big[\forall\, v \in \mathcal{V}:\ (1-\epsilon)\|v\|_2^2 \le \|Sv\|_2^2 \le (1+\epsilon)\|v\|_2^2\Big] \ge 1-\delta.$$

The key point is that the sketching matrix is drawn independently of the subspace. In the least-squares context, the relevant subspace is the column space of A (or the augmented space spanned by the columns of $[A\ b]$), and the OSE property guarantees that the geometry of that subspace is approximately preserved after compression.

The squared-norm convention is the cleanest one for later spectral statements, because it is equivalent to controlling the Gram operator $U^\top S^\top SU$ on an orthonormal basis U of the embedded subspace. It immediately implies the unsquared estimate

$$\sqrt{1-\epsilon}\,\|v\|_2 \le \|Sv\|_2 \le \sqrt{1+\epsilon}\,\|v\|_2,$$

so no geometric content is lost by working with squares.

The OSE condition can be interpreted as a uniform Johnson–Lindenstrauss property on subspaces. This connection is made precise by the Johnson–Lindenstrauss lemma, which provides the foundational concentration result.

Lemma 1.2 (Johnson–Lindenstrauss). Let $\epsilon \in (0,1)$ and let $\mathcal{P}$ be a finite set of p points in $\mathbb{R}^n$. If $S \in \mathbb{R}^{s\times n}$ has entries drawn independently from $\mathcal{N}(0,1/s)$ and

$$s \ge \frac{4\log p}{\epsilon^2/2 - \epsilon^3/3},$$

then with probability at least $1-1/p$,

$$(1-\epsilon)\|u-v\|_2^2 \le \|S(u-v)\|_2^2 \le (1+\epsilon)\|u-v\|_2^2$$

for every pair $u, v \in \mathcal{P}$.

The JL lemma is a statement about finite point sets, but the extension to continuous subspaces follows by a net argument. A γ-net of the unit ball in a d-dimensional subspace has cardinality at most $(3/\gamma)^d$; applying JL to this net and using the Lipschitz continuity of the sketching operator yields the subspace embedding property. The net argument is the standard bridge from pointwise concentration to uniform geometric control.

1.3 Gaussian Sketches

The first sketch family considered in this thesis is the Gaussian sketch. Its entries are independent normal random variables with variance $1/s$, so that

$$S_{ij} \sim \mathcal{N}(0, 1/s).$$

This scaling is chosen so that the sketch is isotropic in expectation. In particular, for a fixed vector $x \in \mathbb{R}^n$, one has the identity

$$\mathbb{E}\|Sx\|_2^2 = \|x\|_2^2,$$

which explains why Gaussian sketches are a natural baseline in both theory and computation.

The main theoretical advantage of the Gaussian model is that the concentration of quadratic forms under Gaussian matrices is well understood. The following result is the central embedding guarantee.

Proposition 1.3 (Gaussian OSE). Let $S \in \mathbb{R}^{s\times n}$ be a Gaussian sketch with entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For any fixed matrix $U \in \mathbb{R}^{n\times d}$ with orthonormal columns and any $\epsilon \in (0, 1)$,

$$\Pr\Big[\|U^\top S^\top SU - I_d\|_2 > \epsilon\Big] \le 2d \cdot \exp\Big(-\frac{s(\epsilon^2 - \epsilon^3)}{4}\Big).$$

In particular, if $s \ge 4(d \log d + \log(2/\delta))/\epsilon^2$, then S is a (d, ϵ, δ)-OSE.

Proof (Proof sketch). For a fixed unit vector $u \in \mathbb{R}^n$, the random variable $\|Su\|_2^2$ is a sum of s independent χ_1^2/s variables. By standard concentration inequalities for chi-squared random variables (the Laurent–Massart bound),

$$\Pr\Big[\Big|\|Su\|_2^2 - 1\Big| > \epsilon\Big] \le 2\exp\Big(-\frac{s(\epsilon^2 - \epsilon^3)}{4}\Big).$$

To pass from a single direction to the entire subspace, cover the unit ball of $\mathrm{col}(U)$ by a γ-net of cardinality at most $(3/\gamma)^d$ with $\gamma = \epsilon/2$. A union bound over the net, combined with the Lipschitz property of the operator norm, yields the stated inequality. The dimension requirement follows by setting the failure probability to δ and solving for s. □

The logarithmic dependence on the ambient dimension n is absent: the embedding quality depends only on the intrinsic dimension d and the sketch size s. This dimension-free character is the theoretical reason why Gaussian sketches are powerful regardless of how large the original problem is.

Closely related are dense sign sketches, often called Rademacher sketches, whose entries are independent $\pm 1/\sqrt{s}$. They retain much of the same isotropic behavior while replacing Gaussian sampling by a simpler discrete distribution. The OSE guarantee for Rademacher sketches is essentially identical to Proposition 1.3→p.3, with the same sketch-size requirement up to constants; the proof uses the Hanson–Wright inequality instead of chi-squared concentration.

1.4 The Subsampled Randomized Hadamard Transform

The third family relevant here is the subsampled randomized Hadamard transform (SRHT). Structured sketches of this type combine random signs, a fast orthogonal transform, and subsampling. They are attractive because they preserve much of the embedding behavior of dense sketches while allowing substantially faster sketch formation when the data layout permits fast transforms.

The SRHT is defined as follows. Let $n = 2^p$ for some integer p. Denote by $H_n \in \mathbb{R}^{n\times n}$ the normalized Walsh–Hadamard matrix (satisfying $H_n H_n^\top = I_n$), and let $D \in \mathbb{R}^{n\times n}$ be a diagonal matrix whose diagonal entries are independent uniform random signs. Let $R \in \mathbb{R}^{s\times n}$ be a matrix that selects s rows uniformly at random without replacement. The SRHT sketch is then

$$S = \sqrt{\frac{n}{s}}\, R\, H_n\, D.$$

The Hadamard matrix can be applied in $O(n \log n)$ operations via the fast Walsh–Hadamard transform, which makes the total cost of forming SA equal to $O(nd \log n)$ instead of the $O(nds)$ cost of a dense Gaussian sketch. The tradeoff is that the SRHT introduces additional combinatorial structure that makes the analysis more delicate.

With this normalization the SRHT is isotropic in expectation after random row selection, so that $\mathbb{E}\|Sx\|_2^2 = \|x\|_2^2$ for fixed x. This is the structured analogue of the Gaussian scaling discussed earlier.

Proposition 1.4 (SRHT Embedding). Let $S \in \mathbb{R}^{s\times n}$ be an SRHT sketch. For any fixed matrix $U \in \mathbb{R}^{n\times d}$ with orthonormal columns and any $\epsilon \in (0, 1)$,

$$\Pr\Big[\|U^\top S^\top S U - I_d\|_2 > \epsilon\Big] \le \delta$$

provided $s \ge C\, \frac{d \log n}{\epsilon^2} \log\!\Big(\frac{d \log n}{\epsilon^2 \delta}\Big)$ for a universal constant C.

The dependence on $\log n$ is the price of the structure: the SRHT needs a slightly larger sketch than the Gaussian model to achieve the same embedding quality, but the gain in sketch-formation time can be substantial. For this thesis, the contrast between Gaussian and SRHT sketches is essential. Gaussian sketches are the cleanest theoretical baseline, and SRHT-type sketches represent the first genuinely structured option whose computational appeal may become decisive for large problems. The SRHT also plays a methodological role: because it is deterministic once the random signs and row indices are fixed, it is easier to reason about in a mixed-precision setting where the randomization and the arithmetic perturbations should be kept conceptually separate.

1.5 Residual and Solution Guarantees

The subspace-embedding property leads directly to approximation guarantees for the sketched least-squares problem. The following result is the basic accuracy statement for the sketch-and-solve paradigm.

Proposition 1.5 (Sketched LS Residual Bound). Let $A \in \mathbb{R}^{n\times d}$ have full column rank, let $x_\star = \arg\min \|Ax - b\|_2$, and let $\hat{x} = \arg\min \|SAx - Sb\|_2$. If S is a $(d+1, \epsilon, \delta)$-OSE with $\epsilon < 1$, then with probability at least $1 - \delta$,

$$\|A\hat{x} - b\|_2^2 \leq \frac{1+\epsilon}{1-\epsilon} \|Ax_\star - b\|_2^2,$$

and hence

$$\|A\hat{x} - b\|_2 \leq \sqrt{\frac{1+\epsilon}{1-\epsilon}} \|Ax_\star - b\|_2.$$

Proof (Proof sketch). Decompose $b = Ax_\star + r_\star$ where $r_\star$ is the optimal residual. Write $\hat{x} = x_\star + h$. The embedding property on the subspace $\mathrm{col}([A\ r_\star])$ gives

$$(1-\epsilon)\|Ah + r_\star\|_2^2 \leq \|S(Ah + r_\star)\|_2^2 \leq (1+\epsilon)\|Ah + r_\star\|_2^2.$$

Since $\hat{x}$ minimizes $\|S(Ax - b)\|_2$, the optimality of $\hat{x}$ for the sketched problem implies

$$\|SA\hat{x} - Sb\|_2^2 \leq \|SAx_\star - Sb\|_2^2 \leq (1+\epsilon)\|r_\star\|_2^2.$$

Combining the two inequalities and solving for $\|A\hat{x} - b\|_2$ yields the stated bound. □

The residual guarantee is natural but not the only quantity of interest. In many applications, one also cares about the solution error $\|\hat{x} - x_\star\|_2$. A bound on the solution error requires additional control over the geometry of the problem, specifically the extent to which the residual contributes to the solution perturbation.

Remark 1.6 (Solution error versus residual error). Residual guarantees do not automatically imply a comparably small perturbation of the minimizer. In the inconsistent case, the sensitivity of $x_\star$ is additionally filtered through the conditioning of A and by the size of the optimal residual relative to the solution itself. This is why the numerical experiments track both residual ratio and relative solution error. The residual bound is the robust first statement, while a formal solution-error theorem requires more careful assumptions than are imposed here. ⌟

These observations together explain the experimental expectations: residual quality stabilizes quickly as s grows, while solution error depends on both ϵ and $\kappa_2(A)$. The transition from residual-dominated to solution-error-dominated behavior is one of the main features that the numerical experiments are designed to capture.

1.6 Conditioning of the Sketched System

Even when the sketch provides a good embedding in exact arithmetic, the numerical behavior of the reduced problem depends critically on the conditioning of SA. The following result connects the embedding quality to the spectral properties of the compressed operator.

Proposition 1.7 (Conditioning of the Sketched Matrix)**.** Let $A \in \mathbb{R}^{n\times d}$ have full column rank with singular values $\sigma_1 \geq \cdots \geq \sigma_d > 0$, and let S be a (d, ϵ, δ)-OSE. Then with probability at least $1-\delta$,

$$\sqrt{1-\epsilon}\,\sigma_d \leq \sigma_{\min}(SA) \leq \sigma_{\max}(SA) \leq \sqrt{1+\epsilon}\,\sigma_1,$$

and therefore

$$\kappa_2(SA) \leq \sqrt{\frac{1+\epsilon}{1-\epsilon}}\,\kappa_2(A).$$

This is a direct consequence of the embedding property applied to the left and right singular vectors of A. The important qualitative message is that a good subspace embedding cannot dramatically worsen the condition number: if ϵ is small, the sketched system is only slightly less well-conditioned than the original.

However, the proposition assumes exact arithmetic. In finite precision, the situation is more nuanced. The computed sketch $\mathrm{fl}(SA)$ is a perturbation of the ideal SA, and the effective condition number of the computed system may be larger than $\kappa_2(SA)$ by an amount that depends on the precision of the sketch-forming arithmetic and on the dimensions n and s. This is precisely where the interaction between randomized approximation and floating-point perturbation becomes important, and it motivates the mixed-precision analysis in ?? $\to$ p.??.

In practice, the condition number of SA is a useful diagnostic. If $\kappa_2(SA)$ remains bounded as s varies, the sketch is preserving the spectral structure of the original problem; if it grows sharply, the sketch dimension may be insufficient, or the sketch may be failing to embed the relevant subspace. The numerical experiments in ?? $\to$ p.?? track $\kappa_2(SA)$ as a function of s for exactly this reason.

1.7 From Embedding to Solver

The sketching matrix is only one part of the algorithm. Once the reduced operator SA has been formed, one still has to solve the compressed least-squares problem in a numerically sensible way. This is where conditioning re-enters the story. A good embedding should preserve not only residual geometry but also enough spectral structure that the reduced problem remains well behaved.

In practical randomized least-squares solvers, two related paradigms appear repeatedly. The first is sketch-and-solve, in which one directly solves the compressed problem. The second is sketch-and-precondition, in which the sketch is used to build a preconditioner for a more accurate iterative method or refinement procedure. Blendenpik is one of the landmark examples: it combines randomized mixing with a high-quality dense least-squares solve and shows that randomized preprocessing can improve performance without abandoning numerical linear algebra discipline [5]. More recent work goes further and shows that randomized least-squares solvers can be designed to be forward stable, and in some formulations even backward stable, rather than merely fast on average [6] [7].

This stability question is especially important for the present thesis. It is not enough to know that a sketch preserves the least-squares objective in exact arithmetic. One also needs to know how the conditioning of SA, and later of any preconditioned or refined system, interacts with the finite-precision arithmetic discussed in ??→p.??. In that sense, modern RandNLA has moved beyond the initial question of whether randomization can accelerate least squares; it now asks under what conditions randomized solvers can be trusted as serious numerical methods [8] [2].

1.8 Sketch-and-Precondition

The sketch-and-solve viewpoint is conceptually clean, but it is not always the best numerical endpoint. A second paradigm uses the sketch to build a preconditioner for the original problem rather than treating the reduced problem as the final approximation. In broad terms, one computes a factorization of the compressed matrix SA, uses that factorization to define a right preconditioner R^{-1}, and then solves the transformed problem

$$\min_y \|AR^{-1}y - b\|_2, \qquad x = R^{-1}y,$$

typically by an iterative method or by refinement around a high-quality residual computation.

This shift in viewpoint matters because the sketch no longer bears the entire burden of solution accuracy. Its role is to improve the conditioning of the operator seen by the downstream solver. If SA captures the geometry of the column space well enough, then the preconditioned matrix AR^{-1} can become substantially better behaved than the original least-squares system. Blendenpik is the classical example of this philosophy: randomized mixing and sampling are used to construct a preconditioner, but the final method remains firmly tied to stable least-squares solving rather than replacing it by a one-shot compressed surrogate [5].

At the same time, recent analysis shows that the numerical details of this paradigm matter more than one might expect. Meier, Nakatsukasa, Townsend, and Webb show that the commonly used form of sketch-and-precondition can lose numerical stability on ill-conditioned problems, even though the underlying sketch may be geometrically accurate in exact arithmetic [9]. Their analysis makes a crucial distinction between having a good randomized preconditioner in principle and embedding that preconditioner into a floating-point algorithm in a stable way. In particular, they advocate solving the preconditioned least-squares problem with an unpreconditioned iterative method applied to the transformed system, rather than relying on the numerically fragile form of the standard algorithm.

This observation sharpens the role of conditioning in the thesis. The object of interest is not merely whether $\kappa_2(SA)$ is moderate, but whether the entire solver architecture built around SA respects that favorable conditioning once rounding errors enter. It is precisely this distinction that motivates the later mixed-precision experiments. A lower-precision sketching stage may be acceptable if the resulting preconditioner still places the downstream method in a benign regime; it is unacceptable if the arithmetic perturbation destroys the conditioning improvement that the randomized step was supposed to provide.

For the present thesis, this paradigm is important for two reasons. First, it clarifies that randomized least squares is not exhausted by direct sketch-and-solve methods; preconditioning is equally central to the modern literature. Second, it provides a natural bridge to mixed precision. Recent work by Carson and Daužickaitė studies precisely this issue for sketching-based preconditioners combined with GMRES-based iterative refinement, showing how separate precision choices in the sketching and factorization stages affect whether the preconditioner remains effective [10]. This is exactly the kind of coupled approximation-and-arithmetic question that later experiments should address.

1.9 A First Experiment

The initial computational study in this thesis is deliberately simple. A random matrix $A \in \mathbb{R}^{n\times d}$ is generated together with a reference solution $x_\star$, and the right-hand side is taken as either

$$b = Ax_\star$$

in the consistent case or

$$b = Ax_\star + e$$

in the noisy case, where e is a perturbation vector. The current Octave prototype uses Gaussian sketches, while the R package extends the same benchmark structure to Rademacher sketches and the SRHT. For each sketch size s, several independent sketches are drawn, the sketched least-squares problems are solved, and the following quantities are recorded:

- the mean absolute error of the averaged solution,
- the residual ratio relative to the unsketched least-squares solution,
- the relative solution error with respect to a reference solve,
- the condition number of the sketched matrix SA.

This experiment plays an important organizational role. Mathematically, it gives a concrete setting in which subspace-embedding heuristics can be compared against observed residual and solution behavior. Computationally, it provides a common benchmark across the current prototype implementations, which already cover Gaussian, Rademacher, and SRHT sketches. In that sense, it is the smallest setting in this project where geometric approximation, conditioning of the compressed operator, and measured computation are meant to line up exactly.

1.10 Why This Experiment Matters

Even before formal guarantees are introduced, the experiment highlights the main questions that drive the later chapters. How large must the sketch dimension s be before the residual becomes reliably close to optimal? How does the conditioning of the compressed operator SA change as s varies? Which sketch family gives the best compromise between simplicity, structure, and numerical behavior? Once mixed precision is introduced, the same framework also makes it possible to ask which arithmetic stages tolerate lower precision and which do not.

The main topics to be developed further are therefore clear: subspace embeddings as the geometric core of sketching, Gaussian and structured sketches as competing algorithmic choices, residual and solution guarantees for sketch-and-solve methods, and the conditioning and finite-precision behavior of the reduced systems. Those topics place the present thesis within the broader RandNLA literature while keeping the attention on the concrete least-squares problems that the code and experiments in this repository already target.

At the same time, least squares is not the final mathematical endpoint of the thesis. It is the first model problem in which sketching, conditioning, and finite precision interact transparently. In the later covariance-transport chapter, the same perturbation viewpoint is transferred from sketched least-squares operators to empirical covariance operators and the matrix functions built from them. The role of the present chapter is therefore foundational: it supplies the geometric and numerical language that the dependence-correction analysis will reuse in a more application-aware setting.

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